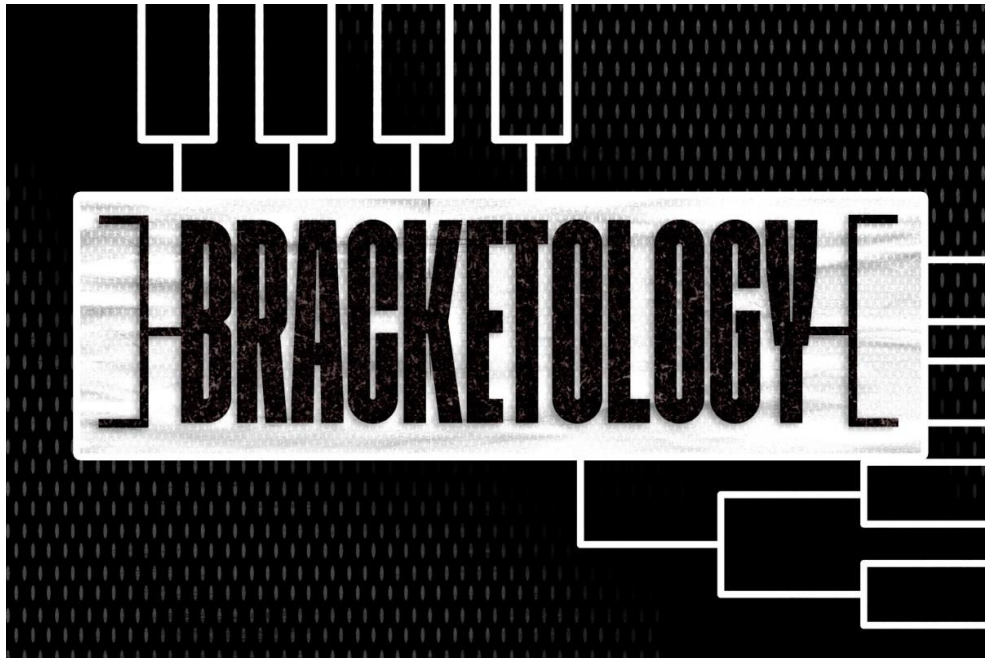
Scenario Trees

Created By Billy Fryer

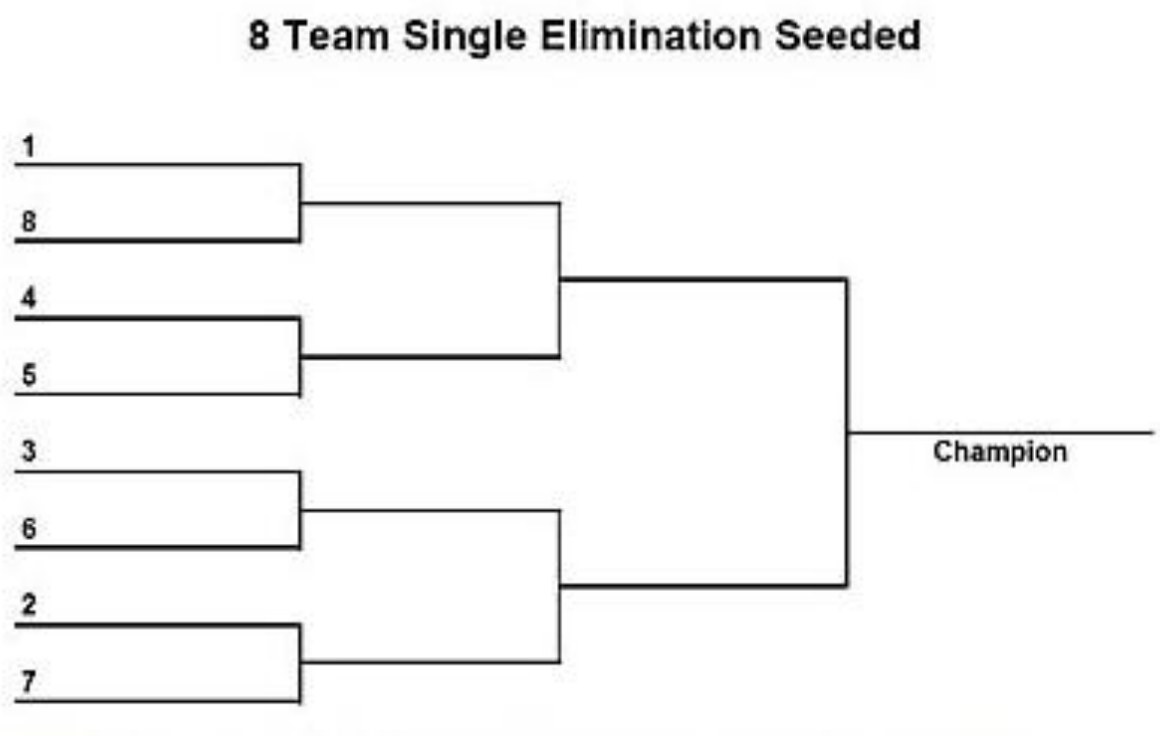


CLINCHED
2024 WNBA PLAYOFFS

Who are we playing?

WNBA Playoff Format (2024)

- Top 8 Teams regardless of conference make the playoffs
- Single Elimination Tournament based off regular season seeding



“Current” WNBA Standings/Relevant Info

- Most teams have 3 games left, Indiana has 2 games left
- New York and Minnesota has clinched the 1 & 2 seeds respectively
- Connecticut, Vegas, and Seattle are all between 2 and 3 games better than Indiana
- Phoenix is more than 2 games worse than Indiana

Relevant Info in Bracket Form

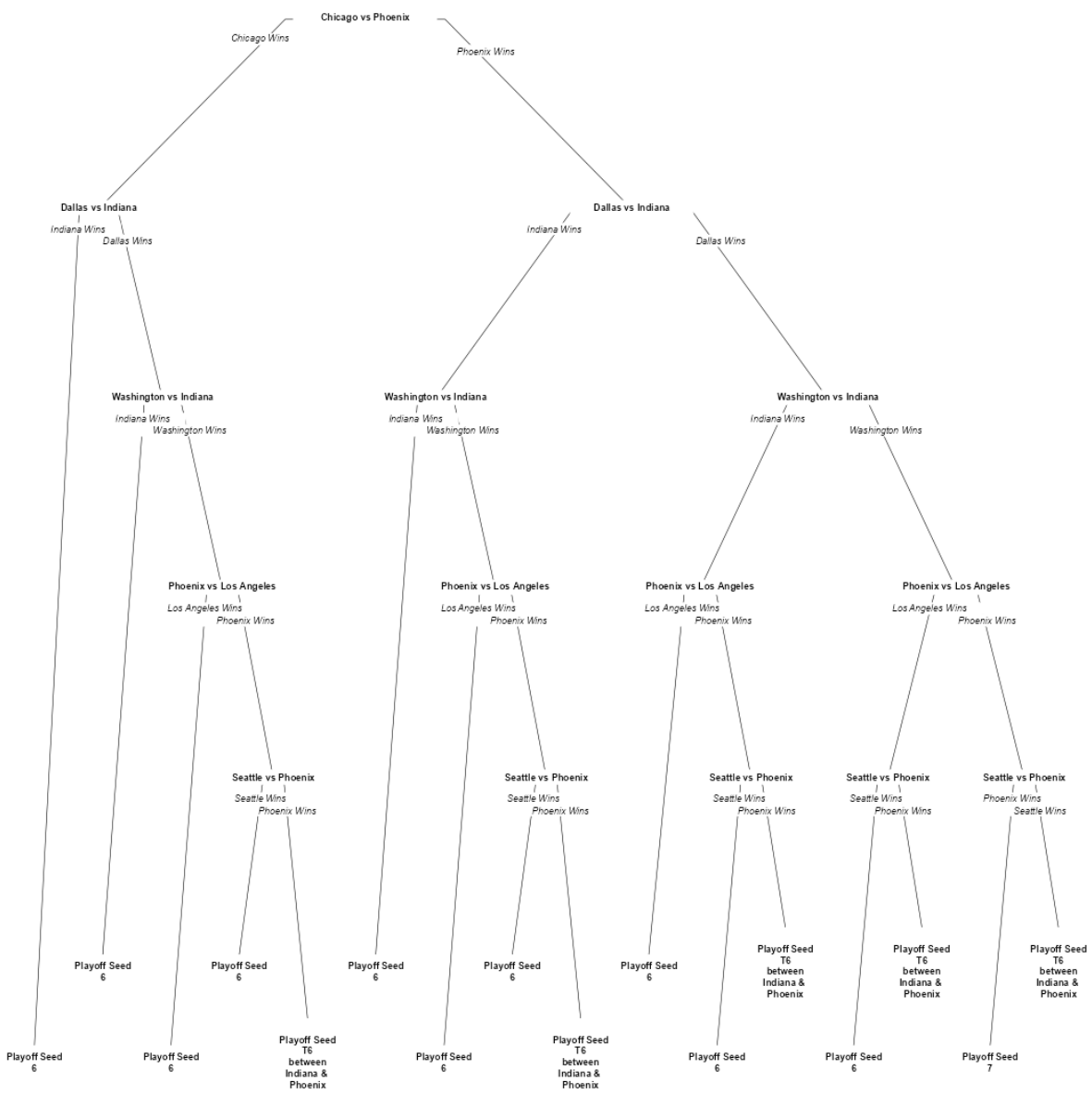
Other Clinched Playoff Teams

What are the possible paths to playing each of these opponents?

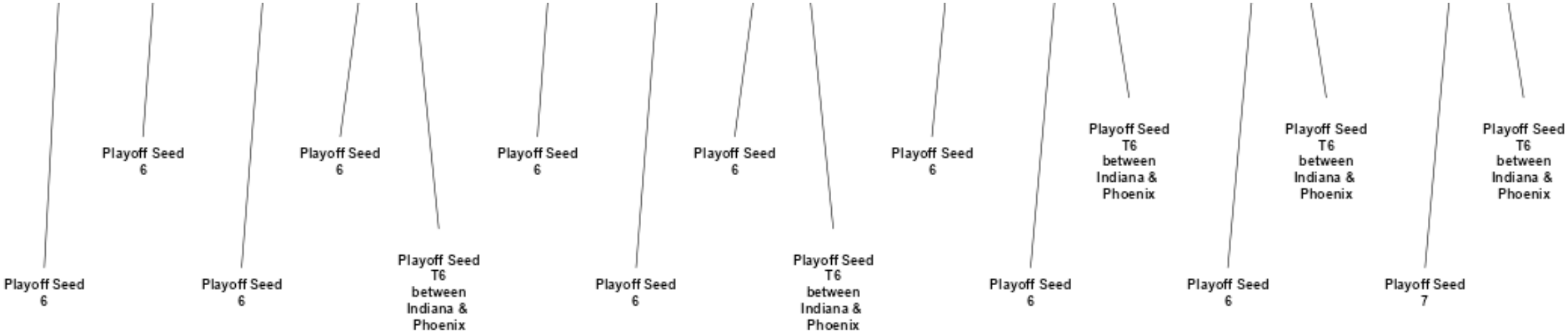
Scenario Trees

- How to Understand Scenario Trees
- How Scenario Trees are Made
- Seed Projection Scenario Trees
- Why is this Important?

Indiana Fever Potential Seeds (Games through 9/13/2024)



Indiana Fever Potential Seeds (Games through 9/13/2024)


Playoff Seed 6
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Playoff Seed T6 between Indiana & Phoenix
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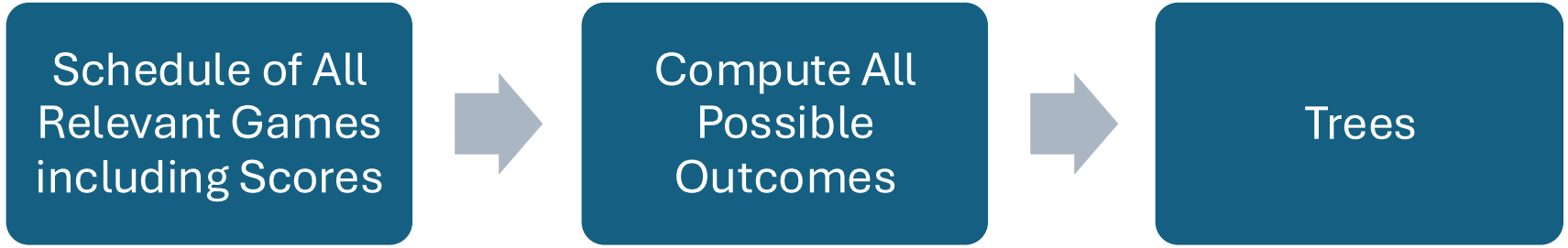
Playoff Seed 6
Playoff Seed T6 between Indiana & Phoenix
Playoff Seed 7
Playoff Seed T6 between Indiana & Phoenix

How It’s Made

- Game Theory – The Original Concept of “Scenario Trees”
- Combinatorics – What are all the possible combinations of results possible?
- Statistics – How can we actually compute these scenarios?

The (Simplified) Process



How The Math Works

- Starting from the completed games, a current standings is calculated
- The games that have yet to be played are thrown into a separate data frame.
- From there, all possible outcomes of games yet to be played are calculated
- Mathematically, if there are *n* games to be played, there are 2^n scenarios because each game has 2 opponents, and only one can win
  - In some sports, there are more than 2 outcomes. Those sports typically go of a points table scoring system, so not relevant here.
- After all the scenarios are enumerated, standings for each scenario are calculated then combined with the standings for the games already played

Games Through 2/25/2026

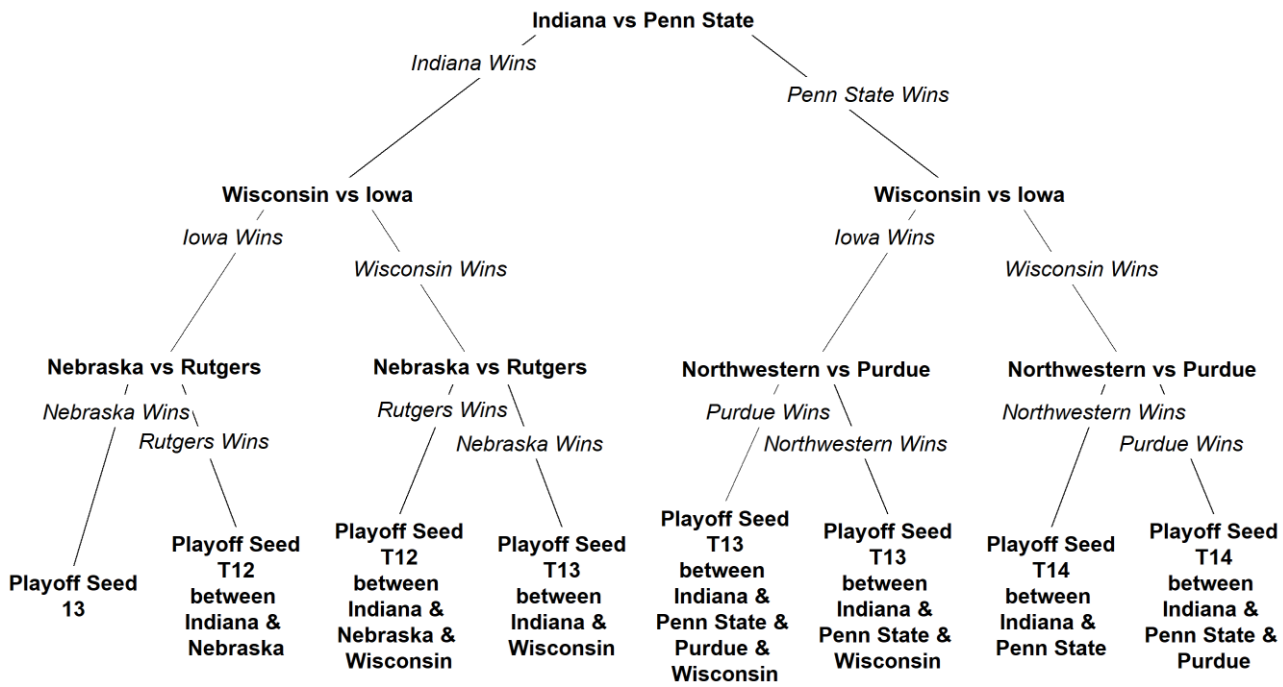
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH
scenario	Iowa vs Illi	Illinois vs N	Indiana vs	Rutgers vs	Wisconsin	Michigan v	Maryland v	Ohio State	Michigan S	Nebraska v	Northwest	Purdue vs (	Oregon vs '	Penn State	USC vs UC	UCLA	Iowa	Michigan	Ohio State	Michigan S	Minnesota	Maryland	Illinois	USC	Washingtc	Oregon	Nebraska	Wisconsin	Indiana	Purdue	Penn State	Northwest	Rutgers
1	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Oregon Wi	Penn State	USC Wins	1	T2	T2	4	T5	T5	7	T8	T8	10	11	12	13	T14	T14	16	17	18
2	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Oregon Wi	Penn State	UCLA Wins	1	T2	T2	4	T5	T5	7	8	T9	T9	11	12	13	T14	T14	16	17	18
3	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Washingtc	Penn State	USC Wins	1	T2	T2	4	T5	T5	7	T8	T8	T8	T11	T11	13	T14	T14	16	17	18
4	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Washingtc	Penn State	UCLA Wins	1	T2	T2	4	T5	T5	7	T8	10	T8	T11	T11	13	T14	T14	16	17	18
5	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Iowa Wins	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Oregon Wi	Penn State	USC Wins	1	2	3	4	T5	T5	7	T8	T8	10	11	12	T13	T13	T13	16	17	18
6	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Iowa Wins	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Oregon Wi	Penn State	UCLA Wins	1	2	3	4	T5	T5	7	8	T9	T9	11	12	T13	T13	T13	16	17	18
7	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Iowa Wins	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Washingtc	Penn State	USC Wins	1	2	3	4	T5	T5	7	T8	T8	T8	T11	T11	T13	T13	T13	16	17	18
8	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Iowa Wins	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Northwest	Purdue Wi	Washingtc	Penn State	UCLA Wins	1	2	3	4	T5	T5	7	T8	10	T8	T11	T11	T13	T13	T13	16	17	18
9	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Purdue Wi	Purdue Wi	Oregon Wi	Penn State	USC Wins	1	T2	T2	4	T5	T5	7	T8	T8	10	11	12	T13	15	T13	16	T17	T17
10	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Purdue Wi	Purdue Wi	Oregon Wi	Penn State	UCLA Wins	1	T2	T2	4	T5	T5	7	8	T9	T9	11	12	T13	15	T13	16	T17	T17
11	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Purdue Wi	Purdue Wi	Washingtc	Penn State	USC Wins	1	T2	T2	4	T5	T5	7	T8	T8	T8	T11	T11	T13	15	T13	16	T17	T17
12	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Wisconsin	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Purdue Wi	Purdue Wi	Washingtc	Penn State	UCLA Wins	1	T2	T2	4	T5	T5	7	T8	10	T8	T11	T11	T13	15	T13	16	T17	T17
13	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Iowa Wins	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Purdue Wi	Purdue Wi	Oregon Wi	Penn State	USC Wins	1	2	3	4	T5	T5	7	T8	T8	10	11	12	T14	T14	13	16	T17	T17
14	Iowa Wins	Illinois Wir	Indiana Wi	Rutgers Wi	Iowa Wins	Michigan V	Maryland V	Ohio State	Michigan S	Nebraska V	Purdue Wi	Purdue Wi	Oregon Wi	Penn State	UCLA Wins	1	2	3	4	T5	T5	7	8	T9	T9	11	12	T14	T14	13	16	T17	T17

Scenario Number (Arbitrarily Assigned)

32,768 Possible Results of the 15 Games Remaining

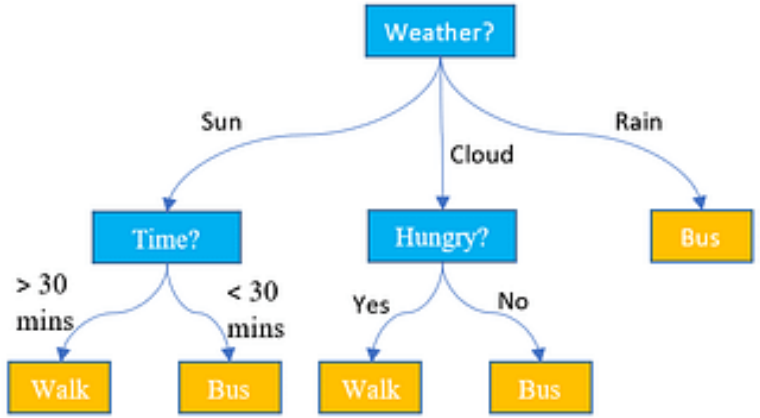
Resulting Seed by Team

Indiana Scenario Tree



How The Trees Works

- Classification Decision Trees
  - Make Rules
  - Rules split the data roughly into even halves
  - Keep splitting until the Complexity Parameter is met
- Complexity Parameter
  - An error threshold where we say “good enough”
  - Typically used to prevent overfitting



How The Trees Works

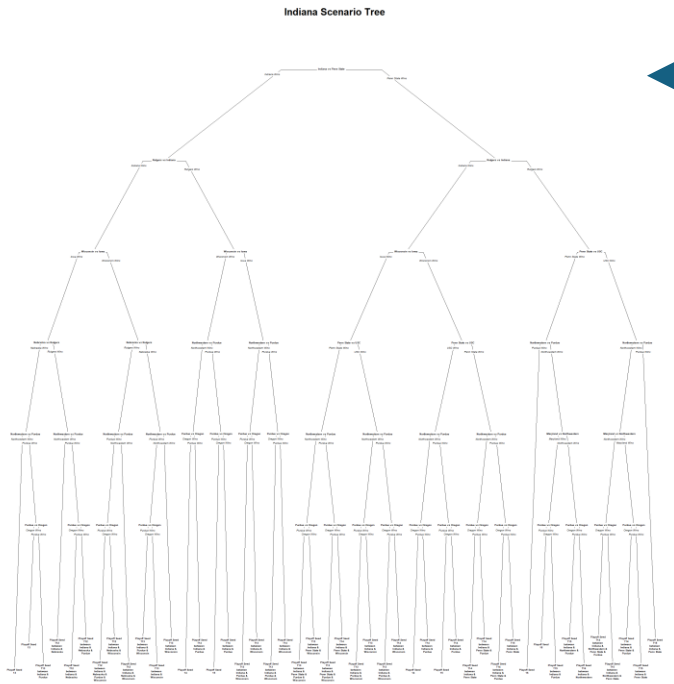
- Overfitting = Good?
  - We have perfectly enumerated data
  - There is no “test dataset”
  - We *do* want to catch all the weird scenarios
    - Who doesn’t want a 7-way tie!?
- Reduce the Complexity Parameter to Essentially 0

Decision Tree Drawbacks

- Max Depth for Decision Trees in R is 30
  - In theory, the furthest out we could project is 30 games remaining in the entire league
  - In reality, that's plenty
    - 30 games left is $2^{30} = 1{,}073{,}741{,}824$ combinations which would brick your computer
    - Really anything over 15 games is hard to visualize with a tree and requires the download of the all-combinations spreadsheet which will also brick your computer

Decision Tree Drawbacks

- Splits are based on Most Important Games (getting the 50/50 on the split algorithm for Decision Trees), not how they occur chronologically

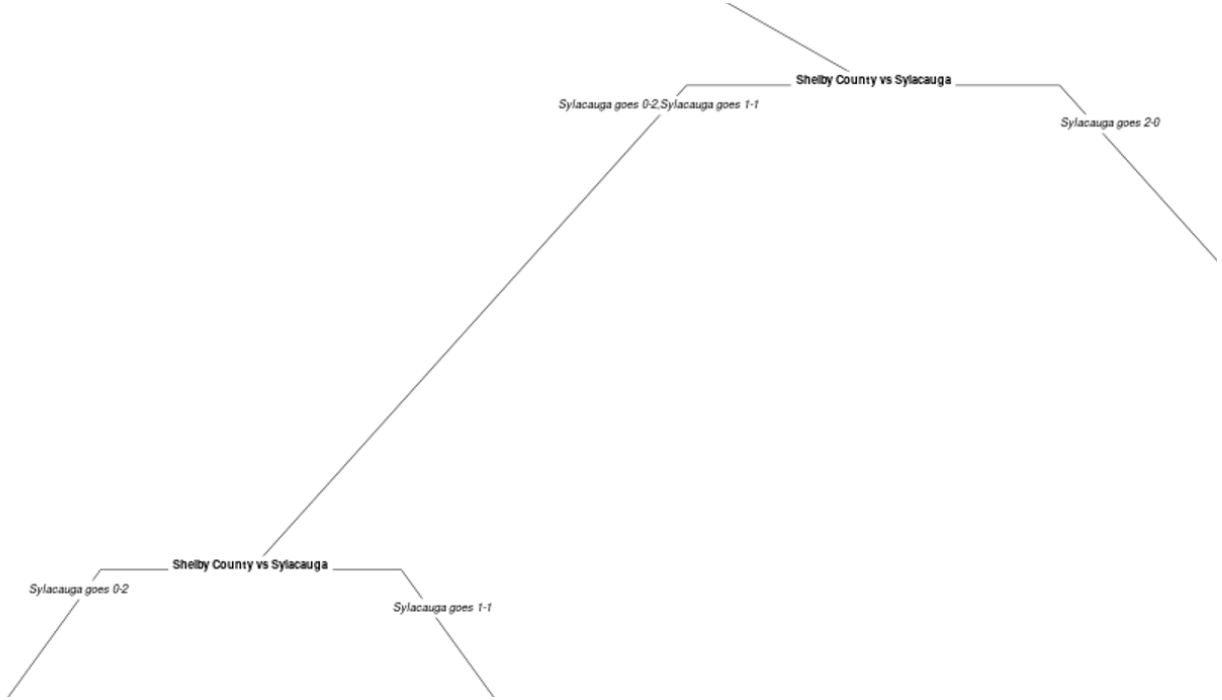


Game is Sunday

Game was Yesterday

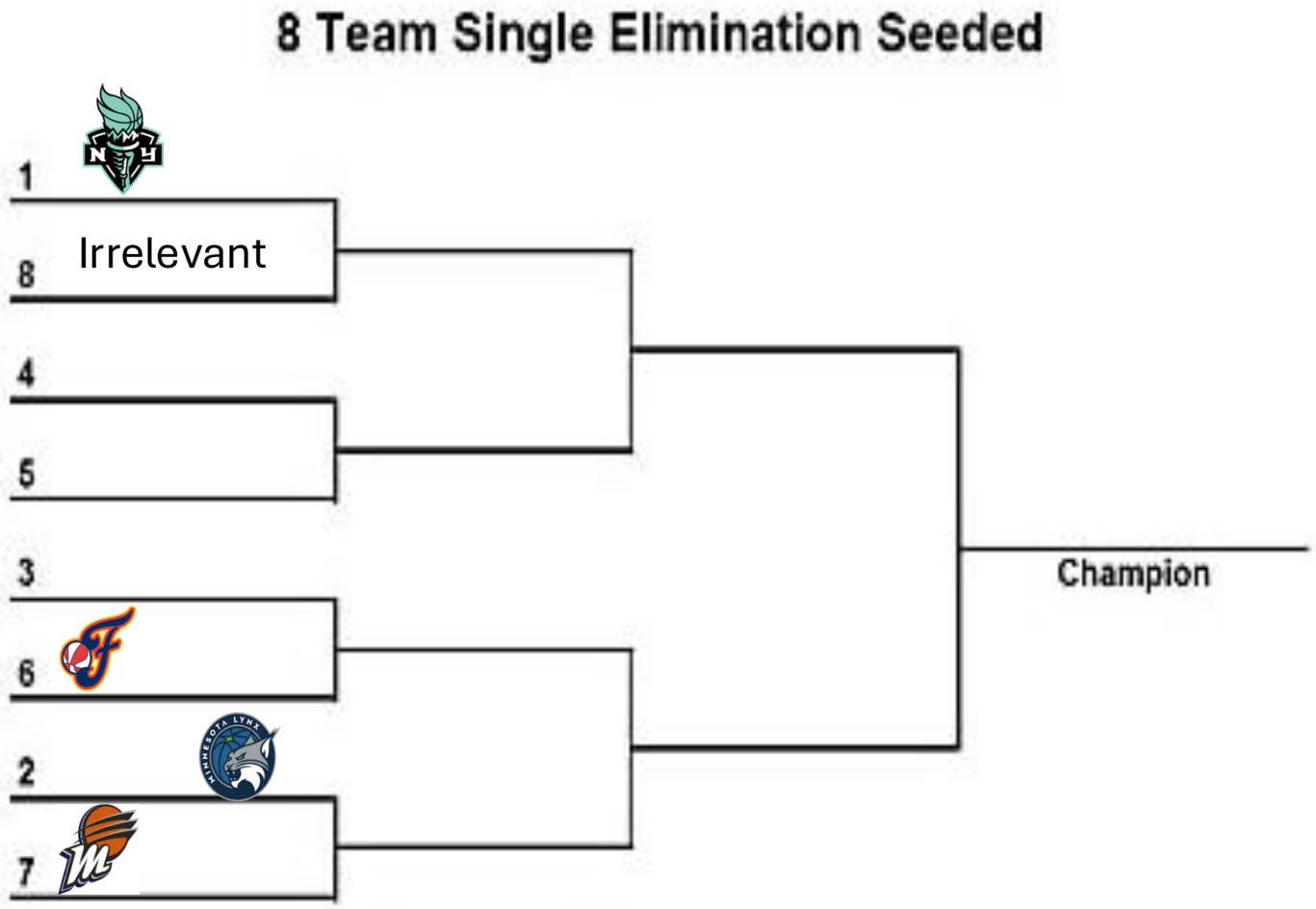
Decision Tree Drawbacks

- Decision Tree splits are Binary
  - What if 2 teams have 2 games remaining against one another?
  - Still possible, the splits get tiered into multiple branches (which unfortunately sometimes get separated

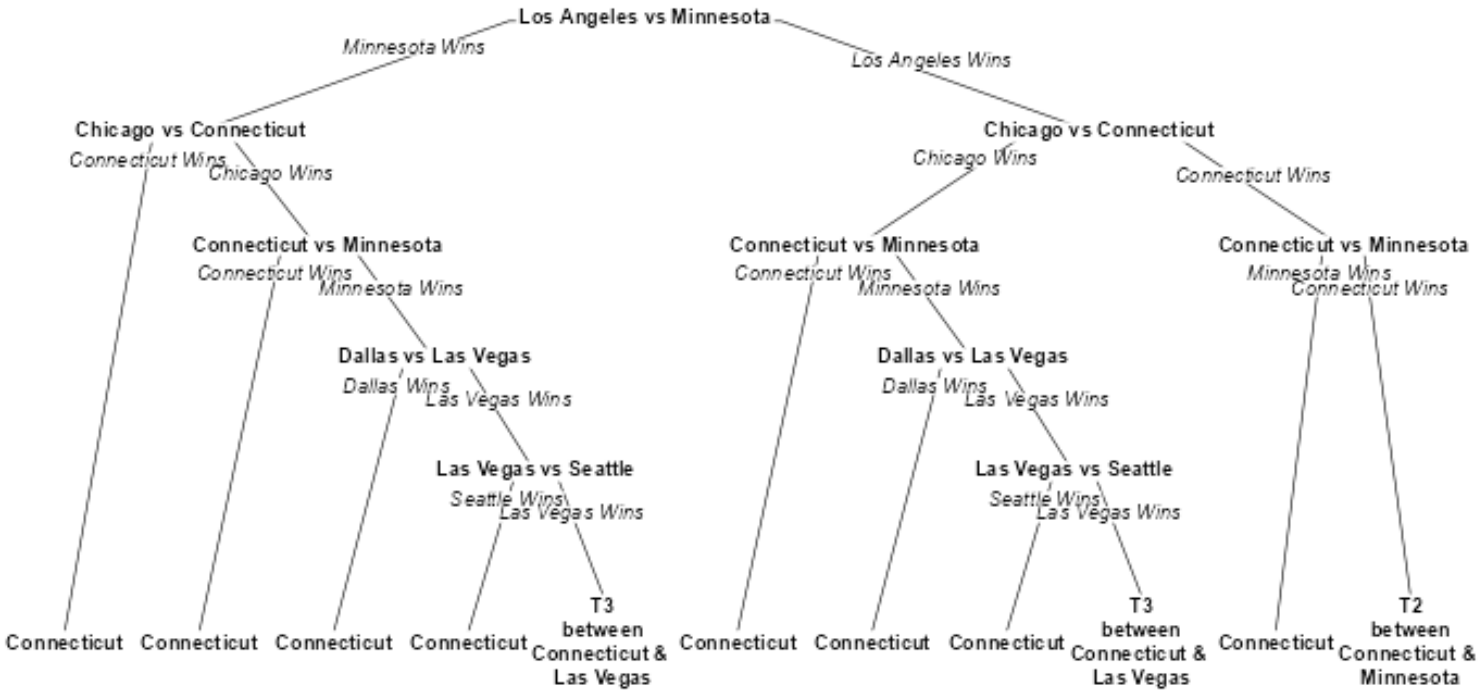


Other Types of Scenario Trees

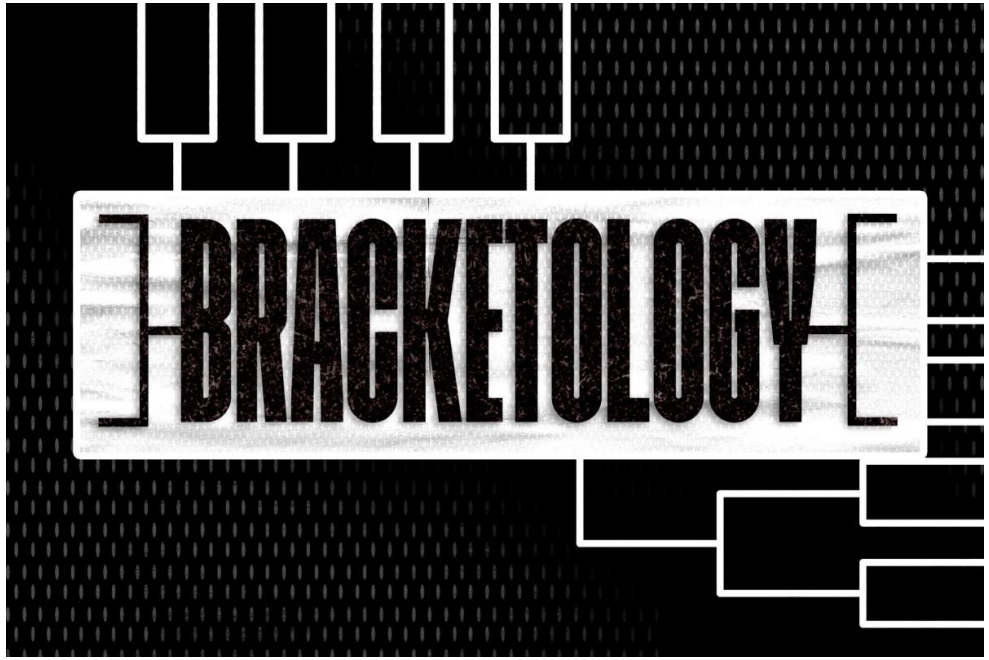
- Predicting Matchups
- We know Indiana is locked in as the 6 seed, who is likely to be the 3 seed?



Potential 3 Seeds (Games through 9/15/2024)



Why Does This Matter?

Game	Time
Indiana vs Washington	6:00 pm
Minnesota vs Los Angeles	6:00 pm
Connecticut vs Chicago	8:00 pm
Las Vegas vs Dallas	8:00 pm

Questions